

Candidate Name	Index Number	CT Group
---------------------------	-------------------------	---------------------

RAFFLES JUNIOR COLLEGE

Preliminary Examinations

**H2 BIOLOGY
PAPER 3**

**9747/3
16th September 2008**

1 hour 30 minutes

Additional material: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers **ONLY in the spaces provided** on the question paper for questions 1 to 3.

Write your answer for question 4 on the separate answer paper provided.

At the end of the examination, **hand in your essay SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Question	
1	
2	
3	
4	
Total	

This document consists of **12** printed pages.



Raffles Junior College
Internal Examination

[Turn over

Answer **all** the questions.

- 1 Fig.1.1 shows an insulin molecule. It is composed of two polypeptide chains: an A chain of 21 amino acids and a B chain of 30 amino acids. These two chains are held together by disulfide bonds formed between two amino acid residues.

For
Examiner's
Use

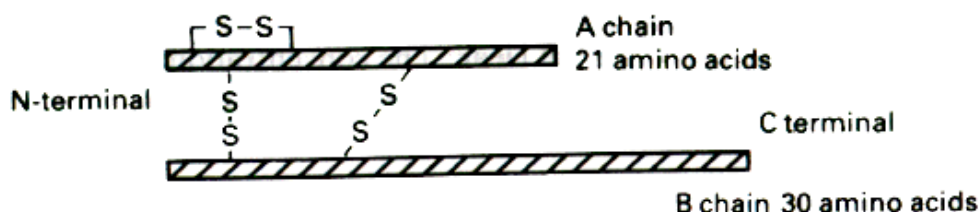


Fig.1.1

- (a) Currently, the production of recombinant insulin involves artificially synthesising the genes for the A chain and the B chain and then inserting each into 2 separate plasmids such that two recombinant plasmids are constructed (Fig.1.2), one carrying the gene for the A chain and one the gene for the B chain.

The A chains and B chains are then synthesised separately in two different mixtures of *E.coli*. The two protein chains are then extracted and purified. Finally, the A and B protein chains are mixed under special conditions allowing disulphide bonds to form *in vitro*.

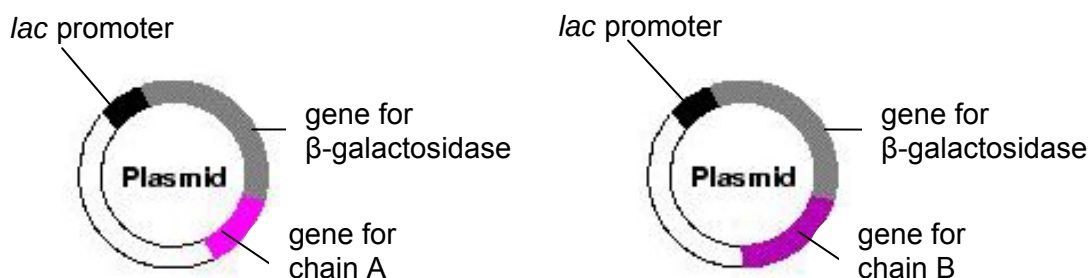


Fig.1.2

- (i) State the function of the β -galactosidase gene in the plasmids.

It functions as a selectable/ genetic marker [1]

- (ii) Why is the *lac* promoter added in both recombinant vectors?

The *lac* promoter, being prokaryotic in origin, would enable more efficient gene expression (compared to a eukaryotic promoter,); [1] as it would be recognised by the host cell which is also prokaryotic in origin [1]

- (iii) Explain how the cells must be treated for transformation to occur.
The bacterial cells must be first made competent (i.e. rendering them capable of taking up DNA from a solution);
by the addition of calcium ions in the form of calcium chloride solution (1);

The cells are then subjected to brief heat shock treatment (a sudden period of heat). This makes pores appear transiently in the bacterial cell membranes, allowing the recombinant DNA molecules to enter the cells.(1)

- (iv) In what ways would the above introduction of recombinant DNA molecules into host cells be similar to mutations?

1) Both introduce new genetic variability into the cell (1);
2) In both cases only one gene or small segment of DNA representing a small fraction of the entire genome is changed or added to the genome.
(1)

- (v) Suggest an advantage of treating diabetics with human insulin produced by genetic engineering.

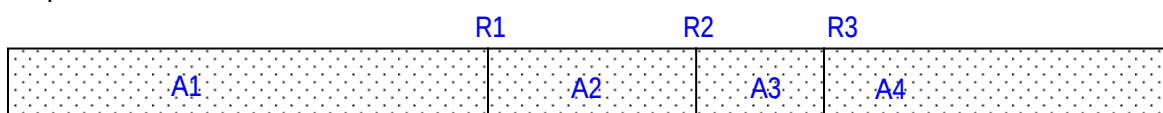
Insulin synthesised this way is

identical to human insulin/pure/uncontaminated and less likely to trigger a possible immune/allergic response (1) ;

- (b) Restriction enzymes can be used to cut DNA at specific sites called restriction sites. Genes can be cut from the chromosome of one species and joined to the chromosome of another species forming recombinant (hybrid) DNA.

Fig.1.3 shows two chromosomes from different species. The specific restriction sites (**R1**, **R2** and **R3**) and the fragments resulting from the restriction digests (**A1**, **A2**, **A3**, **A4**, **B1**, **B2**, **B3** & **B4**) are shown.

Species A



Species B

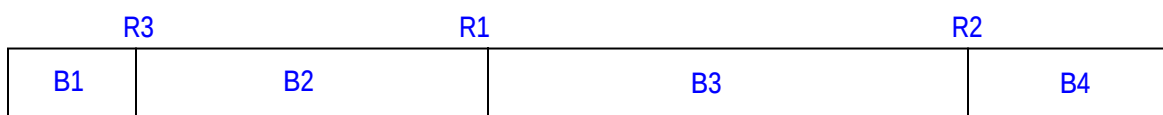
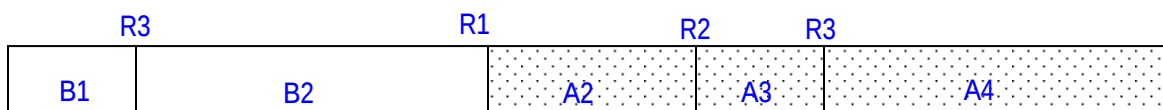
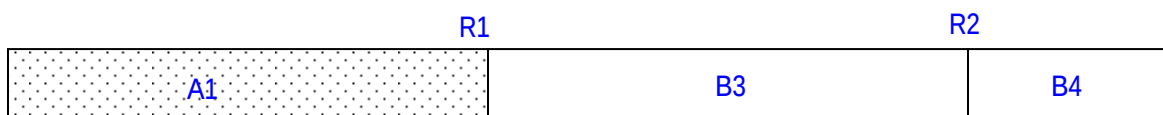
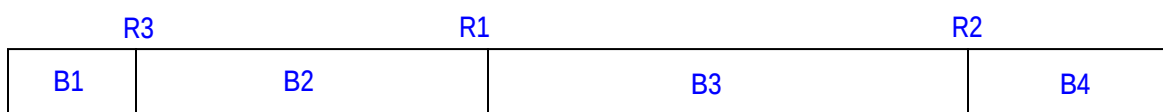
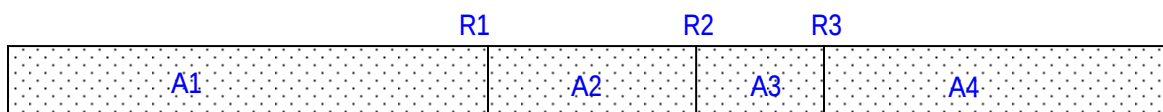


Fig.1.3

Both chromosomes were cut at restriction site R1. The resulting fragments with sticky ends were mixed and allowed to join to form recombinant (hybrid) DNA. In the space below, make labelled sketches of all the possible ways in which the fragments could join. Label all fragments clearly.



1 mark for 2 originals and 1 mark for 2 recombinants, will mark the first 4 and cancel the rest.

[Turn over

- (c) A eukaryotic gene has "sticky ends" produced by the restriction endonuclease *EcoRI*. The gene is added to a mixture containing *EcoRI* and a bacterial plasmid containing an ampicillin resistance gene and a tetracycline resistance gene. The plasmid has a recognition site for *EcoRI* located in the tetracycline resistance gene.

This mixture is incubated for several hours, after which DNA ligase is added. The resulting mixture is then added to bacteria growing in nutrient broth. The bacteria are allowed to grow overnight and then are streaked on plates to produce isolated colonies.

Samples of these colonies are then grown in four different media: nutrient broth with ampicillin, nutrient broth with tetracycline, nutrient broth with ampicillin and tetracycline, and nutrient broth without antibiotics

- (i) What are the natural functions of restriction enzymes? (1)

*It is to cleave any foreign DNA that enters the bacterial cell(1)
They protect bacterial cells from viruses.(1) max 1*

- (ii) How does a bacterial cell protect its own DNA from restriction enzymes? (1)

The bacterial DNA has methylated (adenine and cytosine) residues that prevent restriction enzymes(active site) from binding to the DNA. (1)

- (iii) Explain if the recombinant bacteria will grow in ampicillin containing nutrient broth or tetracycline containing nutrient broth? (2)

*The recombinant bacteria would grow in nutrient broth containing ampicillin as the ampicillin resistance gene would still be intact. (1)
The recombinant bacteria would not grow in nutrient broth containing tetracycline as tetracycline resistance gene would have to be disrupted by the insertional inactivation on the tetracycline resistant gene. (1)*

- (iv) Suggest a potential problem that come about when expressing cloned eukaryotic genes in bacteria?(1)

Eukaryotic genes contain introns which prevent their correct expression in bacterial cells which do not have RNA-splicing machinery (1)

Improper folding of protein in prokaryotes occurs as the endoplasmic reticulum and Golgi apparatus are absent in prokaryotes (1)

[Total : 19 marks]

[Turn over

- 2 (a) Restriction digests were performed separately on pure samples of a plasmid (Fig.2.1) using 4 different restriction enzymes, RE1, RE2, RE3 and RE4.

Lanes 1, 2, 3 and 4 of Fig 2.2 show the result of the restriction digests using restriction enzymes, RE1, RE2, RE3 and RE4 respectively. A DNA ladder (with known fragment lengths) was loaded in Lane 5.

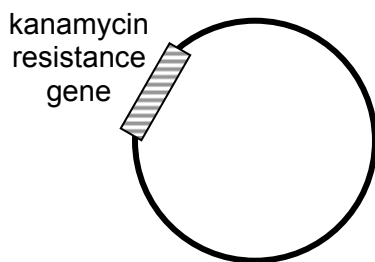


Fig.2.1

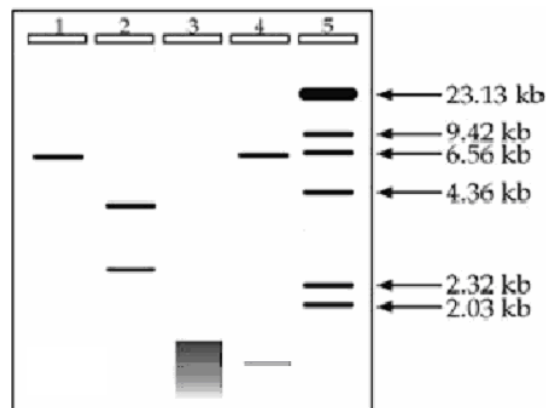


Fig.2.2

- (i) What is the approximate size of the plasmid? [1]
Accept 6 to 6.56 kb (R: anything longer than 6.56kb as shown in lane 1 of the gel electrophoresis)
- (ii) Suggest how these enzymes recognise specific nucleotide sequences. [2]
- **Each type of restriction enzymes has an active site with a specific 3D molecular shape (conformation , not confirmation)**
 - **That corresponds / recognises shape and length of nucleotide sequences at the restriction site**
- (iii) State one limitation in using a plasmid vector [1]
- **There is a limit in the size of the DNA insert that it can carry**
- (iv) If this experiment was performed without any error, give a reason why a smear was obtained in lane 3 [2].
- **RE3 used must have cut the plasmid many times resulting in many smaller fragments**
 - **These fragments are similar in size that they cannot be distinguished between adjacent bands**
- (v) Suggest why this vector is not ideal for use in cloning when we want to insert a foreign gene into *E.coli*. [3]
- **It is not a good vector to be used as it contains only one selectable marker**
 - **At least 2 selectable markers must be present on a plasmid for cloning in this case**
 - **One to allow for selection for transformed bacterial cells that have taken up the plasmid**
 - **The other to select for transgenic bacteria containing gene of interest via insertional inactivation**

For
Examiner's
Use

[Turn over

- (vi) RE4 is known to cut the plasmid only once. However, two bands were seen in lane 4 when the gel was run. Suggest why this is so.

- o Faint band due to supercoiling
- o conatmination

- (b) In a study of a genetic disorder involving one gene, the DNA from six members of a family was isolated, amplified by PCR and run on an agarose gel (Fig.2.3).

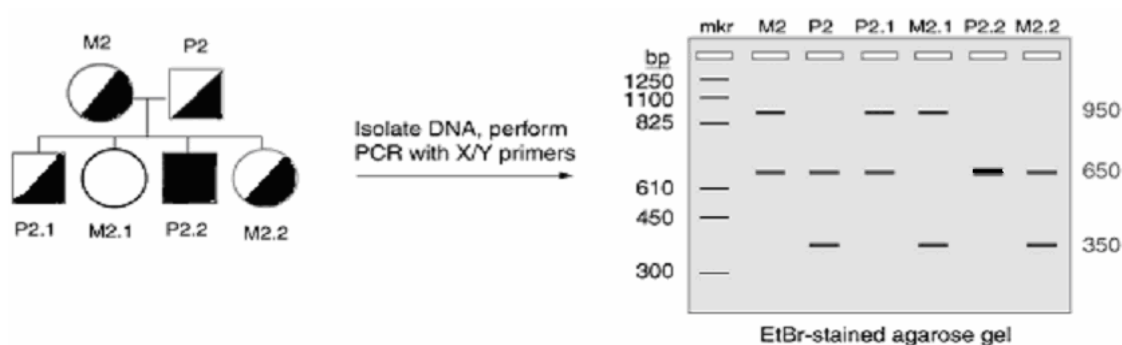


Fig.2.3

The results reveal that there are three forms of the gene (Fig.2.4).

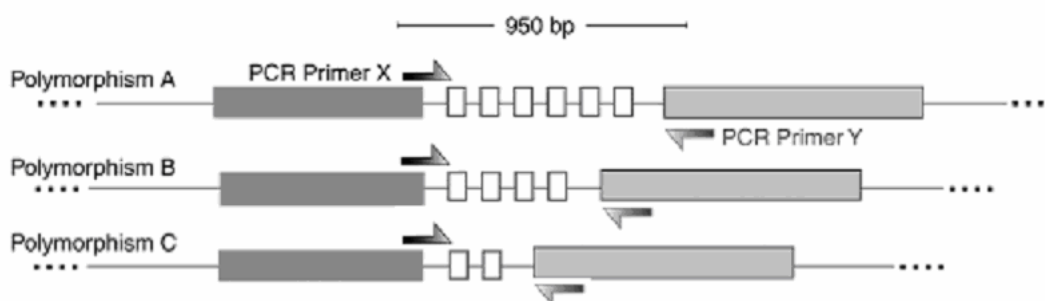


Fig.2.4

- (i) Describe the role of the two different primers in PCR.

*The two different primers mark out the section of the DNA to be amplified by attaching to complementary bases flanking the section of DNA to be amplified. [1]
They provide a free 3'OH end for the attachment and elongation of the DNA polymerase which will add the free dNTPs. [1]*

[Turn over

- (ii) Given that both parents, M2 and P2, are phenotypically normal, which polymorphism marker is associated with this autosomal recessive disease? Justify your answer.

For
Examiner's
Use

The Polymorphism B marker is associated with the disease phenotype [1] because it is the only polymorphism that shows the expected pattern; absent in homozygous normal, one copy in heterozygous and two copies in homozygous disease. [1]

- (iii) What can you conclude if the STRP (Short Tandem Repeat Polymorphism) pattern of one of the offspring was found to be homozygous for polymorphism A?

P2 is not the biological father, as he does not carry the polymorphism A. [1]

[Total : 15]

[Turn over

- 3 Bt crops are so named because they have been genetically altered to produce Bt toxins (called Cry proteins), which kill certain insects. The toxins are naturally produced by the bacterium *Bacillus thuringiensis*, hence the abbreviation Bt.

For
Examiner's
Use

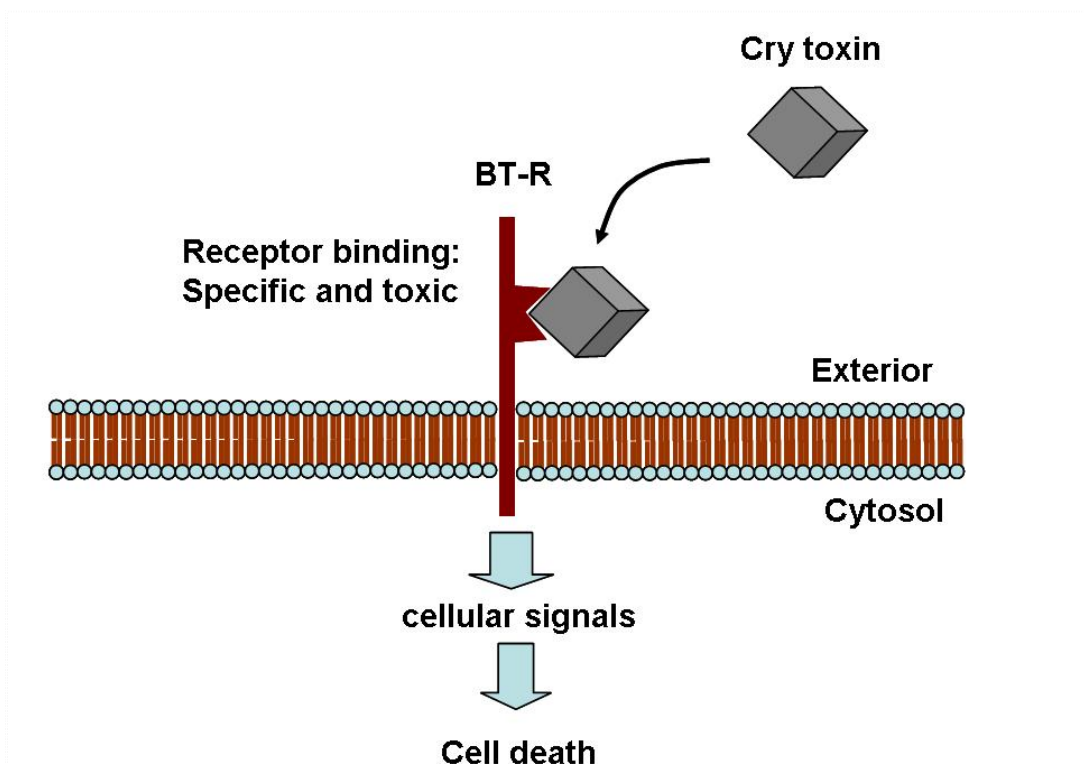


Fig.3.1

- (a) Fig.3.1 shows the proposed model for Cry toxin action. The specific interaction between Cry toxin and the receptor, BT-R, brings about cell death.

- (i) Describe how the receptor is held in the membrane.

The protein receptor is held in the membrane as it contains:

(i) amino acids with hydrophobic R groups which interact with the hydrophobic hydrocarbon tails/hydrophobic core of the phospholipid bilayer (1)

(ii) amino acids with hydrophilic R groups which interact with the hydrophilic head in the phospholipid bilayer (1)

On the cytoplasmic side of the plasma membrane, some membrane proteins are held in place in the membrane by attachment to the cytoskeleton.(1)

On the extracellular side, the proteins are attached to the fibres in the extracellular matrix.(1)

Max 3

[Turn over

(ii) With reference to the information given, explain why the Bt toxin is not harmful to humans in the same way it is to certain insects.

- There is no receptor / BTR for the cry protein/Bt toxin in humans / or the cadherin receptor in humans does not recognize the Bt toxin,
 - [R:if mention receptor without reference to Bt toxin]
- hence there is no receptor binding / ligand-receptor interaction to initiate signal transduction that leads to cell death ;
- Different relay proteins / signal transduction in human and insects hence different cellular response.

Bt-resistant populations of pink bollworm, *Helicoverpa zea*, were found in more than a dozen crop fields in Mississippi and Arkansas between 2003 and 2006. Scientists knew that the Bt-resistant pink bollworms had a mutant version of the receptor, BT-R.

(iii) Describe how pink bollworms have evolved resistance with respect to the Bt - toxin. [4]

1) Mutations in the gene coding for BT-R receptor arose within the pink bollworm populations;;

2) The mutations resulted in amino acid changes and hence changes to the protein conformation/structure of the receptor / unable to bind Bt toxin;;

3) The presence of the Bt crops, provided a selection pressure;;

4) Individuals that carried this mutation / resistance to Bt-toxin have a selective advantage / are selected for / have higher reproductive fitness/must be in the homozygous recessive state to survive ;;

5) These individuals survived and passed on the mutant alleles of cadherin / that confers resistance to their progeny(1); [R: pass on the trait / gene]

6) Over time, freq of the mutant allele / allele that confers resistance / number of individual with the resistance increases in the population. [R: evolution of resistance occur]

[note: points should be made in this order]

(iv) Explain why it is considered a better option to engineer crop plants to make the Bt-toxin, rather than to spray the plants with the toxin. [1]

- Bt-toxin that is directly sprayed onto the plant can be washed away by rain hence not permanent /is laborious /expensive as reapplication is necessary [R: laborious / expensive without explanation].

OR

- In GE crop the BT gene was integrated hence permanent solution / can be passed on to offspring. [NB: not BT-resistant gene]

[Turn over

- (v) Explain two environmental concerns that have been raised regarding the use of Bt crops. [2]

1) That transgenic maize pollen escape into env and crossbreed with wild type variants / weeds, and that the Bt-gene will be transferred to these plants, producing BT toxin and killing non-target insects / disrupt ecological imbalance / which have selective advantage over the wild type;

[R: selective advantage over the wild type w/o mentioning escape / accidentally released into env]

2) The pollen which may express the Bt toxin, will kill non-target organisms like wild species by spreading in the wind; (e.g. pollen blown into the rivers have been known to affect non-target nymphs of insects)

3) Natural predators of the crop herbivore may be affected by the Bt toxin ingested by the herbivore accumulate / forms secondary metabolite / have less prey to feed on hence ecological imbalance.

4) Selective advantage of BT-resistant insects which over time lead to the ineffectiveness of BT-toxin.

- (vi) Give an example of a commercially grown Bt crop.

Bt corn, cotton and tomato [1]

Animals can also be genetically modified.

- (b) Give an example of such an animal and explain how it solves the demand for food in the world.

Transgenic salmon; [2]
grows 3 times faster than its wild counterparts;
higher feed conversion efficiency than its wild counterparts;
higher environmental tolerance as growth hormone is produced throughout the year due to ocean pout antifreeze gene promoter used in gene construct;

[Total : 15]

Write your answer to this question on the separate answer paper provided.
Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.
Your answer must be set out in sections **(a)**, **(b)** etc as indicated in the question.

For
Examiner's
Use

- 3 **(a)** Using appropriate examples, explain how the unique features of stem cells contribute to their functions in a living organism. [8]

Features (adult stem cells):

1. undifferentiated;
 2. replicate indefinitely;
 3. ensure a constant pool of stem cells;
 4. telomerase activated;
 5. respond to appropriate molecular signals;
 6. multipotent – develop into a particular lineage of cells;
- Rej. generic answers pertaining to repair, replace, differentiate (these are weaved into the specific examples)

Max 2 marks

Example 1:

7. embryonic stem cells;
8. pluripotent – develop into all cell types but not a whole organism/except the placenta (or totipotent definition);
9. divide for growth in mass/size/number of cells in developing embryo;
10. differentiate to form different tissues and organs Rej. different cells;

Max 3 marks

Example 2:

11. central nervous system/neural stem cells;
12. differentiated into neurons and glial cells/oligodendrocytes etc (any 2 eggs);

Example 3:

13. hematopoietic/bone marrow stem cells;
14. differentiate into red blood cells/white blood cells/lymphocytes/platelet-forming cells etc.
(any 2 eggs);
15. due to short life-span of cells;

Example 4:

16. epidermis of skin;
17. replace shed/damaged cells/keratinocytes;
18. differentiate into hair follicles, glands etc (any 2 eggs)

Example 5:

19. epithelium of intestine;
20. replace digested/shed cells;
21. differentiate into Paneth cells, goblet cells, absorptive cells etc (any 2 eggs)

Example 6:

22. any other valid example

[Turn over

(b) Discuss how the human genome project can help parents customise their babies and the possible implications of doing so. [6]

Info from HGP [1] must have both to gain 2 marks.

- One can use the information from Human Genome Project to identify a wide range /genome-wide range;
- of disease causing alleles, alleles that give desirable traits such as height, intelligence, strength endurance etc;

(important to state that it is a genome-wide range, since prior to HGP, other genetic traits would have been known).

How info use to create Designer Babies [1] either one is a necessary point.

- Based on info from HGP, Designer babies could 1) be selected for using Pre-implantation Genetic Diagnosis (PGD)/(anyword to that effect) that are disease-free or have desirable traits;
- Or based on the HGP, designer babies could 2) go through genetic modification for enhancements in desirable traits/removal of undesirable traits discovered through info from HGP;

Possible Benefits (at least 2 points to gain full marks)

- Babies that are born would not have to suffer from genetic defects as those that have disease-causing alleles are not implanted or have their genes modified to correct the disease;
- Designer siblings of children who have genetic diseases may be created to provide matching and healthy stem cells for therapy;
- Determine the sex of the child as this might prevent babies who have diseases like haemophilia and Duchenne Muscular Dystrophy, Fragile X syndrome, only show themselves in male babies;

Ethical concerns (at least 2 points to gain full marks)

- For the selection of embryos to implant in PGD, those that are not selected are discarded or unused which will raise ethical concerns since these embryos have been fertilized.
- Designer siblings may be feel that their existence was for a cure rather than parents wanting a child/feel inferior to the child who is sick
- Choosing the sex of babies can raise ethical concerns especially in societies where boys are valued more highly than girls, tipping the sex-ratio.
- Designer babies with genetical enhancements (cf gene therapy) to add genes for HIV resistance or longevity or a high IQ?
- May lead to a social divide between between those who can afford “designer babies”/designer genome and those that cannot.

For
Examiner's
Use

[Turn over

(c) Discuss the applications of plant tissue culture.

[6]

Reference and discussion made to any 6 of the following points:

1. Micropropagation;
2. Hybrid plants (via protoplast fusion);
3. Haploid plants (via anther and pollen culture);
4. Selection and generation of transgenic plants;
5. Conservation of germplasm;
6. Manufacture of secondary metabolites;
7. Pathogen-free stock plants;
8. AVP;

[max 6]

Answers must show relationship of the technique used with its suitability for the application.

[Total : 20]

For
Examiner's
Use

[Turn over